DE–3975 **11**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2012.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2002 Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL the questions.

1. List out six elements of a computer and communication system.

2. Define the terms "Custom software" and "Packaged software".

3. Differentiate "Fax Boards" from "Dedicated fax machines".

4. Define, "Web Browsers" Give examples.

5. Write down the DO-WHILE construct with an illustration.

6. List any five escape sequences and meaning.

7. Give the general format for one-dimensional array statement.

8. What is a pointer? Illustrate.

9. Define: "Macros". Illustrate.

10. State the syntax for "User defined data types".

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Give a note on Database Software.

12. Write about World Wide Web.

13. List and explain the various types of communication channels.

14. Using IF-ELSE construct, develop a 'C' program to find the sum of odd numbers and even numbers, between 0 and 100.

15. Develop a 'C' program to find whether the given string is a Palindrome or not. (Example for Palindrome is "LIRIL").

16. Write a 'C' program to calculate the factorial of an integer quantity using recursion

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Give a detailed description on "Internet" and its features.

18. Develop a 'C' program to find the sum of a trace, above diagonal and below diagonal of a 4 × 4 square matrix.

19. Write a 'C' program to determine the size of a structure using pointers.

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DE–3976 **12**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2012.

OPERATING SYSTEMS

(2002 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What are the goals of an operating system?

2. What are system calls?

3. Define a process.

4. Define a critical section problem.

5. What is the use of a device controller?

6. Define deadlock.

7. Define multiprogramming.

8. What is page breakage?

9. Define a file.

10. What is a directory?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the features of second generation computers.

12. Explain the uses of monitors.

13. Explain race condution and any one method to resolve it.

14. Explain fixed parititioned memory management.

15. Explain any two page replacement algorithms.

16. Explain file sharing and file system reliability.

SECTION C — (2 ×15 = 30 marks)

Answer any TWO questions.

17. Explain the various categories of system calls.

18. Explain deadlock prevention and deadlock avoidance.

19. Explain in detail segmented memory management scheme.

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DE–3977 **13**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2012.

OBJECT ORIENTED PROGRAMMING AND C++

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Define an object. Give an example.

2. What is meant by polymorphism?

3. What is a scope resolution operator?

4. List the user defined data types available in C++.

5. Explain any one characteristics of a static data member.

6. What are destructors?

7. What is a conditional operator?

8. What is a function call operator?

9. What is meant by multiple inheritance?

10. What is a virtual function?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain the procedure-oriented programming approach.

12. Explain the enumerated data type in C++ by giving an example.

13. Explain the two methods for creating symbolic constants in C++.

14. Explain the use of private member functions with an example.

15. Explain hybrid inheritance by giving an example.

16. Explain the use of this pointer with an example.

PART C — (2×15 = 30 marks)

Answer any TWO questions.

17. Explain the basic concepts in object oriented programming.

18. Explain by giving examples overloading of I/O and pointer operators.

19. Explain sequential imput and output operators on files.

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DE–3978 **14**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2012.

DATA STRUCTURES AND ALGORITHMS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL the questions.

1. What is an Algorithm?

2. Define Complexity time.

3. What is Data Structure?

4. Define Arrays.

5. What is string? Give an example.

6. What is strictly binary tree? Give an example.

7. What is Graph? Give an example.

8. What is radix exchange sort?

9. Define Adjacency Matrix.

10. What is Multiway Search Trees?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Write an algorithm to delete a node from a singly linked list.

12. What is Circular linked list? Give its example and it application.

13. Explain the string manipulations with an example.

14. Explain the Binary Tree representation.

15. Explain the selecting sorting with an example.

16. Briefly discussed about weight balanced trees in search trees.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. List the various operations in STACK. Explain any two operations.

18. Discuss about the pattern matching algorithm.

19. Explain the Heap and Quick sort with an example.

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DE–3979 **15**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE
EXAMINATION, MAY 2012.

OBJECT ORIENTED DBMS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Describe the advantages of DBMS.

2. Define Data independence.

3. What are entity and attributes?

4. What are 'Weak entities'?

5. What is Trigger?

6. List the purpose of IDBC.

7. Define functional dependency.

8. What is an 'Index' in physical database design?

9. Define DTD.

10. What is www?

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. (a) Compare file system with DBMS.

    (b) Discuss the structure of DBMS.

12. Explain ER diagram with an example.

13. Discuss the various Aggregate functions in SOL.

14. Explain the various functions to be considered the physical data base design.

15. Compare parallel and distributed databases.

16. List the features of XML.

PART C — (2 ×15 = 30 marks)

Answer any TWO questions.

17. Explain First the three normal forms with examples.

18. Explain the structure of SQL by grieving examples.

19. Bring out the similarities and differences between OODBMS and ORDBMS.

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DE–3980 **21**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2012.

INTERNET PROGRAMMING AND WEB DESIGN

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL the questions.

1. What is security?
2. What is encryption?
3. Mention the advantages of using CGI.
4. What the advantages of using perl?
5. Write notes on internet programming languages.
6. What are the advantages of using SGML?
7. What are the advantages of using HTML?
8. Write note on web page.
9. Why we need animation?
10. Write note on dynamic graphics.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Discuss about WWW design issues.
12. Write in detail about java in windows.
13. Describe the java script in detail.

14. Discuss about Netscape extensions.

15. Write in detail about the creating Netscape navigator plug-ins.

16. Discuss about the adding graphics to web pages.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain in detail about the developing intranet applications with examples.

18. Describe the Microsoft Internet implementation in detail.

19. Write in detail about the creating a custom integrated applications with multiple protocols.

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DE–3981 **22**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2012.

COMPUTER NETWORKS

(2002 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What is computer network?

2. What is unicasting?

3. Define TDM.

4. Define the three types of switching methodologies.

5. What are the functions of data link layer?

6. What is bit stuffing?

7. Compare any three issues between datagram and virtual-circuit subnets.

8. Write a note on firewall.

9. What are the primitives for a simple transport 'service'?

10. What is RPC?

SECTION B — ($4 \times 10 = 40$ marks)

Answer any FOUR questions.

11. Explain the uses of computer networks on various fields.

12. Explain the error detection and correction codes with examples.

13. Describe the one-bit sliding window protocol.

14. Explain the shortest path routing algorithm with an example.

15. Describe in detail the TCP service model.

16. Explain in detail on quantum cryptography.

SECTION C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. With a neat diagram, explain the functions of seven layers in OSI reference model.

18. Describe in detail the public switched telephone network.

19. Explain in detail on Electronic Mail.

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DE–3982 **23**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2012.

SOFTWARE ENGINEERING

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Define the term real model.

2. What is projection metric?

3. What do you mean by software plan?

4. What is meant by decomposition technique?

5. Write notes on tracking.

6. What are the basic concepts of project scheduling?

7. Write notes on communication techniques.

8. Define the term hierarchy of system engineering.

9. Write notes on architectural design.

10. Define the term data design.

PART B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Discuss about the linear sequential model.

12. Explain the prototyping model in detail.

13. Discuss about the risk projection.

14. Explain the software quality assurance in detail.

15. Explain the specification modeling and Information flow in detail.

16. Describe the black box testing in detail.

PART C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. What are the tools to measure software and the description about formal methods model? (15)

18. Discuss about the software risk identification. (15)

19. Explain the ISO 9000 quality standards in detail. (15)

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DE–3983 **24**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2012.

VISUAL PROGRAMMING

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)
Answer ALL questions.

1. Name any three components of window of specify their purpose.
2. What is a DCC?
3. What is a property window?
4. Specify the purpose of a form.
5. Give the syntax of input box.
6. What is the need for a database system?
7. What is an event?
8. Define the term socialization.
9. Expand OLE, DLL, ODBC.
10. What is an exception?

PART B — (4 × 10 = 40 marks)
Answer any FOUR questions.

11. Bring out the differences between traditional and windows programming.

12. Explain :

    (a) Device context

    (b) SDK tools.

13. What is meant by event driver programming? Explain.

14. Explain input box and msg box with example.

15. Write note on serialization.

16. Explain OLE with examples.

PART C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Discuss the creation of a form with properties and event procedures.

18. Discuss the database interaction features of VB with examples.

19. Discuss:

    (a) Document view architecture.

    (b) VC ++ components.

DE–3984 **25**

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, APRIL 2012.

MULTIMEDIA AND APPLICATIONS

(2002 onwards)

Time : Three hours Maximum : 100 marks

PART A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What do you understand about multimedia?

2. Write short notes on commercial Tools.

3. Define Analog/Digital audio.

4. Write short notes on speech audio.

5. Define JPEG.

6. Define Redundancy and visibility.

7. Define Format classes.

8. Define objects.

9. Define media types.

10. Define CD-i.

PART B — ($4 \times 10 = 40$ marks)

Answer any FOUR questions.

11. Explain system components.

12. Discuss multimedia platforms.

13. Explain Temporal and non temporal media types.

14. Explain Video compression techniques.

15. Discuss Media classes.

16. Explain multimedia on networks.

PART C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Explain the applications of multimedia in various fields.

18. Discuss Image compression standards.

19. Explain Multimedia architecture and organization.

DE–4368 **16**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2012.

LAB I — C++ LABORATORY

(2002 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

1. Write a C++ program to find the length of a string and to reverse the same.

———————————————— cut here ————————————————

2. Write a program in C++ to multiply two matrices A of order $m \times n$ and B of order $n \times p$.

———————————————— cut here ————————————————

3. Write a program in C++ to find the transpose of a given matrix A.

———————————————— cut here ————————————————

4. Write a program in C++ using pointers to arrange a given set of numbers in descending order.

———————————————— cut here ————————————————

5. Write a program in C++ to find the length of 2 strings and to concatenate them.

6. Write a program in C++ to calculate simple and compound interest given $P = 2,00,000$, $n = 5$ years and $r = 13\%$.

---------- cut here ----------

7. Write a program in C++ to implement the concept of binary operator overloading.

---------- cut here ----------

8. Write a program in C++ to implement the concept of multiple inheritance.

---------- cut here ----------

9. Write a C++ program to implement the concept of virtual function.

---------- cut here ----------

10. Write a menu driven C++ program to display the number, name and address of the donors for the following categories.

    (a) Blood donors having the blood group AB+

    (b) Blood donors in the age group of 18 to 30

    (c) Female blood donors with blood group A in the age between 15 and 25.

DE–4369 **17**

DISTANCE EDUCATION

M.Sc. (Information Technology) DEGREE EXAMINATION, MAY 2012.

DATA STRUCTURES LABORATORY

(2002 onwards)

Time : Three hours Maximum : 100 marks

Break-up of marks :
Record Note Book : 10
Algorithm/Flow chart : 10
Program : 50
Debugging : 10
Execution : 10
Result : 10
Total : 100

Examiner has to choose TWO questions One from Section A and another from Section B.

SECTION A

1. Write a C++ program to find the transpose of 3×3 matrix with an example.

---------- Cut here ----------

2. Write a C++ program to sort the n elements in ascending order using heap sort.

---------- Cut here ----------

3. Write a C program to sort the n elements in descending order using insertion sort.

4. Write a C++ program to insert and delete data elements in a circularly linked list.

---------- Cut here ----------

5. Write a C++ program to sort the n elements using selection sort in descending order.

---------- Cut here ----------

SECTION B

6. Write a C++ program to insert and delete data elements in doubly linked list.

---------- Cut here ----------

7. Write a C++ program to implement the concept of LIFO in stack with an example.

---------- Cut here ----------

8. Write a C++ program to insert and delete elements in the queue in FIFO order with an example.

---------- Cut here ----------

9. Write a C++ program to search the elements in a group of elements using linear search method.

---------- Cut here ----------

10. Write a C++ program to create a singly linked list and performing its operations using array.

DISTANCE EDUCATION

M.Sc.(Information Technology) DEGREE EXAMINATION, MAY 2012.

LAB III — INTERNET PROGRAMMING LABORATORY

(2002 onwards)

Time : Three hours Maximum : 100 marks

Examiner to select and give ONE question to each candidate.

Record Note Book	:	10
Algorithm	:	10
Program	:	50
Debugging	:	10
Execution	:	10
Result	:	10
Total	:	100

---------- Cut here ----------

1. (a) Write a Java Script that implements String Concatenation.

   (b) Create an Employee Database using JDBC.

---------- Cut here ----------

2. (a) Write a Java program using applet to display a line using different colours.

   (b) Write a Swing Program to create buttons with

      (i) Image

      (ii) Tool Tip.

3. (a) Write a VB Script to do the following :

(i) Calculate the leap year

(ii) Check the string is palindrome or not.

(b) Write a Swing program to create images and buttons.

---------- Cut here ----------

4. (a) Write a Java Script program to create a calculator.

(b) Write a Java program to implement shape using applets.

---------- Cut here ----------

5. (a) Create a Database for Income Tax processing.

(b) Create a HTML document to create a table using table tags.

---------- Cut here ----------

6. (a) Write a swing program to create Label, Text box and image.

(b) Create a Java script program to verify Email address.

---------- Cut here ----------

7. (a) Create a Java script program to verify email address.

(b) Write a Java script program to read the student marks for 5 subject and print the total and average.

---------- Cut here ----------

8. (a) Write a VB script program to scroll the text on status bar.

(b) Write a Java program to generate random numbers using random Class and arrange the number in ascending order and descending order.

9. (a) Write a Java Script that implements String Concatenation.

(b) Design a Web page using HTML code to display the details of Air Reservation form.

DISTANCE EDUCATION

M.Sc. (IT) DEGREE EXAMINATION, MAY 2012.

Lab IV — VISUAL PROGRAMMING LABORATORY

(2002 onwards)

Time : Three hours Maximum : 100 marks

ONE question is to be given to the candidate by lot systems.

1. (a) Write event procedures to perform the following:

   (i) display date and time

   (ii) to reverse the string.

   (b) Write a VC++ program to find out whether a mouse in attached or not

---------- Cut here ----------

2. (a) Using control array, create a calculator with operation mentioned below.

   (i) addition

   (ii) subtraction

   (iii) multiplication.

   (b) Write a VC ++ program to create a list box in a window.

3. (a) Using MS flex grid control, display the multiplication and addition tables of 20 rows and columns.

(b) Write VC ++ program to generate a status box and show the status of caps lock, Num lock in it.

---------- Cut here ----------

4. (a) Create a form in the text box, combo box and command button if do the operations:

(i) add university name in the combo box

(ii) search and delete particular university in the combo box.

(iii) sort university names.

(b) Write VC ++ program to get the status of shift and toggle keys using MFC.

---------- Cut here ----------

5. (a) Using RDA object, implement the operation with employee details.

(i) insertion

(ii) search

(iii) delete &

(iv) update.

(b) Write VC ++ program to fill back ground of the client area with a bit map.

6. (a) Using ActiveX control, create a text box that accepts only numeric value with properties listed below :

   (i) background

   (ii) fore ground &

   (iii) Text.

   (b) Write VC ++ program to handle windows message in MFC program.

---------- Cut here ----------

7. (a) Using data access object create a student application to do the following :

   (i) instruction of records

   (ii) delete the records and

   (iii) update the records.

   (b) Write VC ++ program to create a window of desired size using MFC.

---------- Cut here ----------

8. (a) Develop a data report using Employee details with operations

   (i) display all the employee details.

   (ii) display total salary for each department.

   (b) Write VC ++ program to create a list box in a window.

9. (a) Using Active X DCL or E × E add a class module that performs the functions

(i) Test whether a number is odd or not

(ii) Test whether a number is divisible by 5 and not divisible by 10

(b) Write VC++ program to get the status of the shift and toggle keys using MFC.

---------- Cut here ----------

10. (a) Write event procedure for the following

(i) to reverse a string

(ii) to chedle whether a string in palindrome or not

(iii) to change the case from upper to lower.

(b) Write a VC ++ program to handle windows message in MFC program.